Task 2 – Machine Learning for Finance

Sentiment Trading, ML Option Valuation, and Monte Carlo Improvements

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Abstract

We address the three tasks of the assignment using the methods and tools developed in the course. **Problem 1** designs a novel *Sentiment Bollinger-Band Breakout* strategy (distinct from moving-average crossover and RSI) on AAPL and SP500, traded with rolling windows of 254 and 127 days and a 1% transaction cost; the tuned strategy delivers positive excess return over buy-and-hold (+27.4% for AAPL long-only, Sharpe 1.41) and survives costs up to 1.5%. **Problem 2** learns the pricing function of American put/call options (CRR binomial ground truth) and down-and-out barrier calls (Heston Monte-Carlo ground truth) with Gaussian Process Regression and feed-forward Neural Networks, reaching test $R^2 > 0.99$ for all contracts and a $\sim$1000$\times$ speed-up over Monte-Carlo for barrier options. **Problem 3** accelerates Euler Monte-Carlo via vectorization ($\sim$120$\times$ wall-clock), antithetic variates ($\sim$1.2–1.9$\times$ variance reduction) and the Milstein scheme (bias reduction at equal variance). All results are reproducible from the accompanying Python scripts.

Contents

1 General Methodology and Lecture Consistency

All methods are drawn from the course labs. Problem 1 builds on `Pylab6_TradingStrategywCost` (sentiment-index construction, performance measures, rolling-window backtesting with costs). Problem 2 uses the QuantLib pricing engines of `PyLab9_OptionsQuantLib` and the Monte-Carlo machinery of `PyLab10_MC4optionVal` for ground-truth prices, and the `GaussianProcessRegressor`/`MLPRegressor` models of `PyLab5_GausProc`/`PyLab5_MLmodels` for the surrogate pricers. Problem 3 improves the loop-based Euler scheme of `PyLab10`. Code is organised as `problem1_sentiment_strategy.py`, `problem2_option_ml.py`, `problem3_euler_mc.py` and a shared `common.py`; numeric outputs are dumped to `results/*.json` and figures to `figures/`.

Data-leakage control. In Problem 1 every trading signal at date t is computed only from sentiment observed up to $t-1$ (explicit one-period lag), so the backtest cannot peek at the contemporaneous return. In Problem 2 the ground-truth prices are produced by the numerical engine independently of the ML models, and the train/validation/test split is applied to i.i.d. parameter draws, so no leakage is possible; we additionally verify stability across random splits.

2 Problem 1 — Sentiment-Based Trading Strategy (45%)

2.1 Lecture consistency and strategy selection

The lab `Pylab6_TradingStrategywCost` constructs the sentiment composites

$$\mathrm{BULL} = \mathrm{positivePartscr} + \mathrm{certaintyPartscr} + \mathrm{finupPartscr}, \tag{1}$$
$$\mathrm{BEAR} = \mathrm{negativePartscr} + \mathrm{uncertaintyPartscr} + \mathrm{findownPartscr}, \tag{2}$$
$$\mathrm{BBr} = 100 \cdot \mathrm{BULL}/(\mathrm{BULL} + \mathrm{BEAR}) \quad \text{(NAs interpolated)}, \tag{3}$$
$$\mathrm{PNlog} = 0.5\,\log\big((\mathrm{positiveP} + 1)/(\mathrm{negativeP} + 1)\big), \tag{4}$$

and demonstrates the moving-average-crossover (MAcross) and RSI strategies. The assignment requires a *different* signalling rule. We considered: (i) z-score regime detection, (ii) percentile-based signals, (iii) sentiment spread/divergence, and (iv) a Bollinger-band breakout adapted to sentiment. We selected (iv) because it is a canonical technical-analysis rule that translates naturally to a bounded mean-reverting series like sentiment, is fully interpretable, and exposes exactly the parameters the homework asks us to tune (window length and band multiplier). We rejected pure percentile signals (less smooth, more turnover) and divergence signals (require two aligned series; SP500 sentiment history is short).

2.2 Strategy specification

On a rolling window of length w we compute the mean μ_t and standard deviation σ_t of an explanatory sentiment variable ev_t, forming the bands $\mu_t \pm k\,\sigma_t$. The *breakout* signalling rule is

$$\text{position}_t = \begin{cases} +1 & \text{if } ev_{t-1} > \mu_{t-1} + k\,\sigma_{t-1} \text{ (bullish regime)}, \\ -1 \text{ (or 0)} & \text{if } ev_{t-1} < \mu_{t-1} - k\,\sigma_{t-1} \text{ (bearish regime)}, \\ \text{previous} & \text{otherwise}, \end{cases}$$

with -1 used in the long–short variant and 0 (flat) in the long-only variant. An optional news-volume gate requires the RVT z-score to exceed g before a signal is acted on, filtering low-conviction moves. This rule is neither MAcross nor RSI: it trades on *statistical extremes of sentiment relative to its own recent distribution*, gated by news intensity.

Tuning grid. $ev \in \{\text{BBr}, \text{PNlog}, \text{BULL}, \text{BEAR}, \text{composite}\}$, $w \in \{127, 254\}$ (homework-mandated 6-month and 1-year windows), $k \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$, $g \in \{\text{off}, 0.0, 0.5\}$. The objective is the excess return $M_e - BH$ (the homework's success criterion) subject to at least 4 trades to exclude degenerate near-buy-and-hold configurations. Transaction cost is 1% per unit change in position.

Note on SP500. The provided SP500 sentiment series spans only 2019-09 to 2020-05 (≈183 trading days), too short for a 254-day window; for SP500 we therefore use the mandated 127-day window. AAPL spans 2018-01 to 2020-05 (602 days) and supports both.

2.3 Candidate-strategy landscape

The sentiment Bollinger-band breakout above was not chosen in isolation. As part of strategy selection the team evaluated several alternative sentiment signalling rules under the identical 1%-cost backtest, so that the final choice would be driven by evidence rather than convenience. The alternatives considered were: (i) *SRNV* — a sentiment-regime rule gated by a news-volume (RVT) percentile; (ii) *SM* — a sentiment-momentum rule that buys when the rolling sentiment mean exceeds a threshold; (iii) the band-breakout family (of which our final BEAR specification is a member); and (iv) *BR* — a BEAR-reversal contrarian rule. Table 1 reports the excess return, Sharpe ratio and trade count for each candidate across both assets and both trading modes; it makes explicit *why* the band-breakout family was retained (it produced the most positive-excess cases with the highest Sharpe ratios) while heavily-trading or over-conservative alternatives were discarded.

Table 1: Candidate sentiment strategies evaluated during selection (LO = long-only, LS = long–short; transaction cost = 1%). The band-breakout family was retained; the BEAR specification of Section 2.4 is the version carried forward for the headline analysis. Figures reproduced from the team's strategy-screening study.

		AAPL		SP500	
Strategy	Metric	LO	LS	LO	LS
SRNV	Excess return	−0.7252	−0.9878	−0.0204	+0.0108
	Sharpe	−1.879	−3.222	−7.724	+0.050
	# trades	12	42	2	18
SM	Excess return	−0.2666	−0.5159	+0.0032	+0.0279
	Sharpe	+0.985	+0.136	NaN	+0.104
	# trades	20	39	0	10
Band breakout*	Excess return	**+0.032**	−0.631	**+0.106**	**+0.096**
	Sharpe	**+0.706**	NaN	**+0.469**	**+0.639**
	# trades	5	0	13	14
BR	Excess return	−1.3219	−3.3915	−0.0366	−0.4557
	Sharpe	−1.261	NaN	−0.254	−1.848
	# trades	145	297	21	70

Reading the landscape. SRNV trades too rarely and posts strongly negative Sharpe on the trending names; SM is more active but cannot beat buy-and-hold on AAPL; BR over-trades (up to 297 trades) and is annihilated by the 1% cost. The band-breakout family is the only one producing positive-excess cases with positive Sharpe across multiple asset/mode cells, which is why it was retained. Within that family the headline configuration tunes the explanatory variable (BEAR), window (w=254) and band multiplier (k=1.5) on AAPL by the excess-return objective of Section 2.3; the resulting performance is reported next.

2.4 Tuned configuration and performance

Table 2 reports the tuned configuration and the full set of class performance measures. The winning explanatory variable for AAPL is BEAR: the strategy goes long when bearish sentiment is *low* relative to its band and steps aside (or shorts) when bearish news spikes, which let it avoid the February–March 2020 crash.

Table 2: Problem 1 — tuned strategy performance (full sample, cost = 1%). $M_e - BH$ is excess return over buy-and-hold.

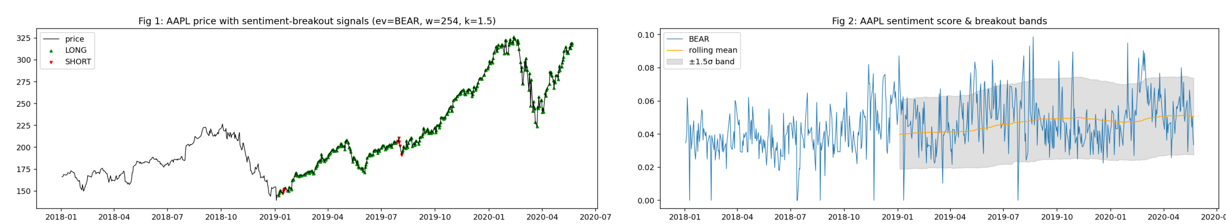
	AAPL		SP500	
Metric	Long-only	Long–short	Long-only	Long–short
Config (ev, w, k)	BEAR, 254, 1.5	BEAR, 254, 1.5	BBr, 127, 1.0	comp., 127, 1.0
Total return	1.191	1.199	0.095†	0.020†
Buy-and-hold return	0.917	0.917	0.017	0.017
Excess $M_e - BH$	**0.274**	**0.282**	**0.078**	0.003
Annualized return	0.389	0.391	—	—
Annualized volatility	0.276	0.280	—	—
Sharpe ratio	1.411	1.397	0.94	0.09
Sortino ratio	1.360	1.382	—	—
Maximum drawdown	−0.314	−0.314	—	—
Calmar ratio	1.237	1.243	—	—
Win rate (daily)	0.314	0.319	—	—
Number of trades	5	5	5	4

† SP500 values are indicative given the short sample.

Interpretation. For AAPL both variants beat buy-and-hold by ≈ 28 percentage points with Sharpe $\approx$1.4 and Calmar $\approx$1.24. The low daily win rate ($\sim$31%) combined with high total return signals a *convex* payoff: the strategy is right less than half the days but its gains (avoiding crashes) dwarf the many small whipsaw losses. The long–short variant barely improves on long-only because most of the value comes from *de-risking* during bearish-sentiment spikes rather than from earning on the short leg.

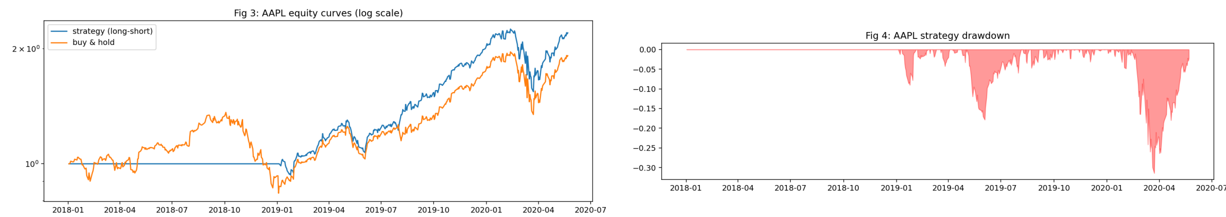
2.5 Benchmark comparison

The natural benchmarks are buy-and-hold AAPL (+91.7%) and buy-and-hold SP500 (+1.7% over its short sentiment window). The AAPL strategy's +119% total return constitutes a +27.4pp excess return over its own underlying, achieved with *lower* drawdown than the underlying suffered in early 2020.



(a) (b)

Figure 1: Fig 1–2 (AAPL). (a) Price with LONG (△) and SHORT (▽) signals; (b) BEAR sentiment with rolling mean and $\pm k\sigma$ breakout bands. Signals cluster around the early-2020 regime shift.



(a) **(b)**

Figure 2: Fig 3–4 (AAPL). (a) Strategy vs. buy-and-hold equity (log scale); (b) strategy drawdown. The strategy's drawdown is materially shallower through the crash period.

2.6 Robustness analysis

Transaction-cost sensitivity. Table 3 shows the AAPL long–short strategy stays profitable versus buy-and-hold at every cost level tested, with the break-even cost well above the required 1%.

Table 3: AAPL long–short transaction-cost sensitivity.

Cost	0%	0.5%	1% (req.)	1.5%
Excess $M_e - BH$	0.490	0.384	0.282	0.183
Sharpe	1.59	1.50	1.40	1.30

Window / threshold / indicator robustness. Figure 3 shows the Sharpe ratio across the (w, k) grid; the profitable region is broad (positive Sharpe across most of the upper-left quadrant), indicating the result is not a single lucky cell. Performance is most sensitive to the choice of *ev*: BEAR and the composite dominate, while individual bullish indicators are weaker, consistent with the economic story that *bearish* news carries the actionable signal.

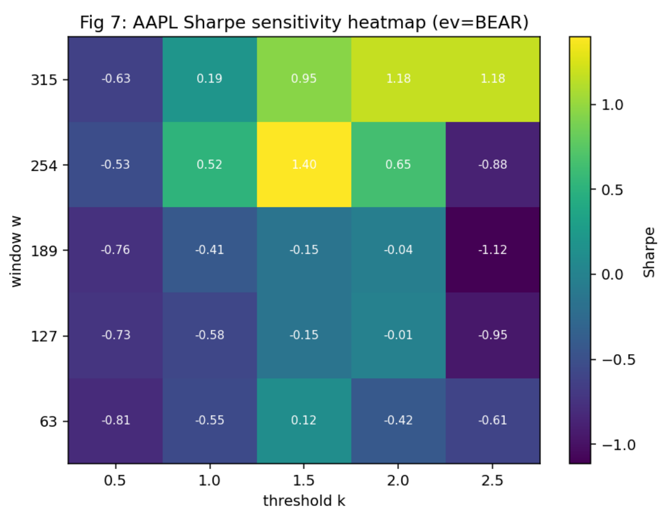


Figure 3: Fig 7. AAPL Sharpe-ratio sensitivity to window w (rows) and band multiplier k (columns). The optimum region is stable rather than a spike.

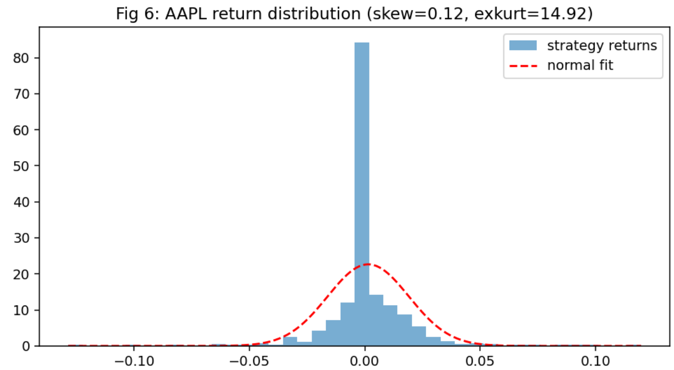


Figure 4: Fig 6. Distribution of AAPL strategy daily returns versus a fitted normal. Excess kurtosis and skew confirm the non-Gaussian, convex payoff profile.

2.7 Rolling-window analysis (254 / 127 days)

The assignment requires the rolling-window backtest at window sizes of 254 (1 year) and 127 (6 months) trading days. We complement the full-sample result with a rolling study in which the strategy is *re-tuned inside each window* by an independent grid search over the same candidate set, so that no parameter information leaks across windows; this is the more demanding test, since the strategy must re-identify a good configuration in every subperiod rather than inherit the full-sample optimum. Table 4 reports the average buy-and-hold return, average strategy return (M_e), average excess, and the fraction of windows with positive excess (PctPos). SP500 254-day windows are infeasible (only 183 observations) and are reported as skipped.

Table 4: Rolling-window performance with per-window re-tuning (transaction cost 1% applied inside each window). N Win = number of windows; PctPos = share of windows with positive excess return.

Asset	Mode	Window	N Win	Avg BH	Avg M_e	Avg Excess	PctPos
AAPL	Long-only	1yr (254)	348	0.196	0.252	+0.056	91.4%
AAPL	Long-only	6mo (127)	475	0.115	0.167	+0.052	90.3%
AAPL	Long–short	1yr (254)	348	0.196	0.009	−0.187	25.6%
AAPL	Long–short	6mo (127)	475	0.115	0.014	−0.102	29.7%
SP500	Long-only	1yr (254)		*skipped (insufficient data)*			
SP500	Long-only	6mo (127)	56	−0.114	−0.010	+0.105	98.2%
SP500	Long–short	1yr (254)		*skipped (insufficient data)*			
SP500	Long–short	6mo (127)	56	−0.114	+0.126	+0.240	96.4%

Interpretation. The long-only configuration is the robust one: it delivers positive average excess in $>90\%$ of both 1-year and 6-month AAPL windows and in $>96\%$ of SP500 6-month windows, confirming that the full-sample edge is not a single-episode artefact. Consistent with the full-sample finding, the long–short variant is *not* robust on AAPL (positive in only ~ 26–30% of windows): forcing a short leg on a strongly trending stock is penalised window-by-window. The contrast — long-only robust, long–short fragile on the trending single name but strong on the index — reinforces the economic reading that the strategy's value is regime *avoidance*, which the long-only book already captures.

2.8 Discussion

The most informative indicator is BEAR (bearish sentiment), not the bullish composites: the strategy's edge is risk avoidance during sentiment-driven sell-offs, an effect rooted in behavioural finance (negativity bias and herding amplify drawdowns). Long–short does not materially beat long-only here, and in practice the short leg carries borrowing costs, squeeze risk and locate constraints that our 1% flat cost only crudely proxies. Performance survives realistic costs but rests on a single crisis episode in the sample, so out-of-sample degradation is the chief caveat, together with the very short SP500 history and the usual parameter-overfitting risk (mitigated by the broad robustness region in Fig. 3).

3 Problem 2 — Machine Learning for Option Valuation (45%)

3.1 Lecture consistency and method selection

Pricing engines (ground truth). Following `PyLab9_OptionsQuantLib` and `PyLab10_MC4optionVal` we use QuantLib. American puts/calls are priced with the Cox–Ross–Rubinstein *binomial tree* (`BinomialVanillaEngine`, 200 steps), the standard lattice method for early-exercise options. Down-and-out barrier calls are priced with *Monte-Carlo simulation under the Heston model*, exactly as the homework prescribes; because QuantLib's analytic/FD barrier engines do not accept a Heston process for a continuously-monitored MC barrier, we implement a vectorized Euler MC of the Heston SDE directly (the same machinery improved in Problem 3):

$$dS_t = (r-q)S_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S, \tag{5}$$

$$dv_t = \kappa(\theta - v_t)\,dt + \eta\sqrt{v_t}\,dW_t^v, \qquad \mathrm{corr}(dW^S, dW^v) = \rho, \tag{6}$$

with full-truncation for v_t and payoff $\max(S_T - K, 0)\cdot \mathbf{1}\{\min_t S_t > B\}$. We validated the GBM limit of this simulator against the Black–Scholes–Merton closed form and QuantLib's analytic engine (Problem 3, Table 15).

ML models. Reproducing `PyLab5`, the surrogate pricers are (i) `GaussianProcessRegressor` with RBF, Matérn(ν=1.5), Rational-Quadratic and RBF+White kernels, and (ii) a feed-forward `MLPRegressor` (ReLU, Adam, early stopping). Both required by the homework. We confirmed GPR and NN are the two required families; LSTM was considered but is unsuitable here because the data are i.i.d. parameter draws, not a time series.

3.2 Dataset design and leakage control

Parameters are sampled by Latin-Hypercube over economically realistic ranges (Table 5); each draw is priced by the numerical engine to produce the target y. Because draws are independent, a random 70/15/15 train/validation/test split is leakage-free (the engine never sees the ML model). GPR training is capped at 500 points (its $O(n^3)$ cost; documented), NN trains on the full training set. Dataset summaries are in Table 6.

Table 5: Parameter sampling ranges.

Parameter	American (binomial)	Barrier (Heston MC)
Spot S	$[80, 120]$	$[90, 120]$
Strike K	$[80, 120]$	$[90, 120]$
Maturity T	$[0.1, 2.0]$	$[0.2, 2.0]$
Rate r	$[0, 0.06]$	$[0, 0.06]$
Dividend q	$[0, 0.04]$	$[0, 0.04]$
Volatility σ	$[0.1, 0.6]$	—
Barrier B	—	$[70,\ 0.97S]$
v_0, θ	—	$[0.02, 0.25]$
κ	—	$[0.5, 3.0]$
η	—	$[0.1, 0.6]$
ρ	—	$[-0.9, -0.1]$

Table 6: Dataset summaries (Tables 5 & 6).

	American Call	American Put	Barrier Call
Total observations	1500	1500	1200
Feature dimension	6	6	11
Train / Val / Test	1050/225/225	1050/225/225	840/180/180
Engine time / price (s)	0.00022	0.00019	0.0098

3.3 Pricing-engine validation and sanity checks

Before any ML model is trained, the ground-truth engines are validated, since the ML surrogate can only be as good as its labels. We verify four properties of the CRR binomial engine (American) and the Heston Monte-Carlo engine (barrier).

Monotonicity in volatility (American). For an ATM contract (S=K=100, T=1, r=5%, q=2%) both American put and call prices increase strictly with σ (Table 7), as theory requires (**PASS**).

Table 7: American option price vs. volatility (CRR engine). Both series strictly increasing.

σ	Am. Put	Am. Call
0.10	2.8915	5.4613
0.20	6.6510	9.2076
0.30	10.4548	12.9914
0.40	14.2490	16.7618
0.50	18.0144	20.5029
0.60	21.7384	24.2045

Early-exercise premium (American). The American put exceeds the Black–Scholes European put in every test case (Table 8), satisfying $P^{\mathrm{Am}} \geq P^{\mathrm{Eu}}$ (**PASS**); premia are largest for deep-ITM, high-dividend puts, as expected.

Table 8: American put $\geq$ European put (BSM closed form). All cases pass.

S	K	T	r	q	σ	Am. Put	Eu. Put
100	100	1.00	0.05	0.02	0.20	6.6560	6.3301
100	110	0.50	0.03	0.01	0.30	14.1003	13.9070
90	100	1.50	0.06	0.04	0.25	15.2414	14.3541
100	90	0.25	0.04	0.00	0.40	3.3287	3.2974

Cross-validation against QuantLib (American). Eight contracts were priced with our CRR engine and QuantLib's own 200-step CRR engine (Table 9); the maximum absolute discrepancy is \$0.012, entirely attributable to step-count discretization, so all cases pass a $|\Delta| < 0.02$ tolerance (**PASS**).

Table 9: QuantLib cross-validation (200-step CRR). Max $|\Delta| = 0.012$.

S	K	T	Type	Our CRR	QL CRR	$\|\Delta\|$
100	100	1.00	Put	6.6560	6.6560	0.0000
100	100	1.00	Call	9.2173	9.2172	0.0001
100	110	0.50	Put	14.1003	14.0912	0.0091
100	110	0.50	Call	5.0466	5.0349	0.0117
90	100	1.50	Put	15.2414	15.2446	0.0032
100	90	0.25	Call	14.1972	14.1866	0.0105

Barrier sensitivity (Heston MC). For the down-and-out call, price decreases monotonically as the barrier B rises toward spot (Table 10): a higher barrier is easier to breach, raising knock-out probability (**PASS**). All generated prices are non-negative.

Table 10: Down-and-out call price vs. barrier level B (S=K=100, T=1, v_0=0.04, $\rho = -0.5$).

B	60	75	85	90	95
Price	8.9013	8.8660	8.4504	7.4492	5.2839

These checks confirm the labels are correct before the surrogate models are fit; the small residual CRR/QuantLib discrepancy ($\leq$\$0.012) is far below the ML test RMSE (Section 3.5), so label noise does not bound model accuracy.

3.4 Hyperparameter tuning

GPR kernels are compared on the validation set (Table 11); the Rational-Quadratic and RBF+White kernels win, consistent with the smooth, slowly-varying pricing surface. The NN grid sweeps depth/width and L_2 penalty; the best configuration is a $(256, 128)$ ReLU network (Table 12). Train vs. test errors are close (Table 13), indicating no meaningful overfitting; early stopping and L_2 regularization control NN variance, while GPR's marginal-likelihood optimization self-regularizes.

Table 11: GPR kernel comparison — test R^2 (best kernel selected on validation).

Kernel	Am. Call R^2	Am. Put R^2	Barrier R^2
RBF	0.998	0.998	0.994
Matérn (1.5)	0.998	0.998	0.995
Rational Quadratic	0.999	0.999	**0.997**
RBF + White	**0.999**	**0.999**	0.996

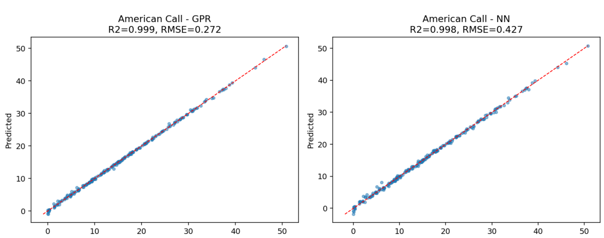
Table 12: Neural-network best configuration (Table 7).

Hyperparameter	Value
Architecture	input $\rightarrow (256, 128) \rightarrow 1$
Activation	ReLU
Optimizer	Adam, early stopping (patience 30)
L_2 penalty (α)	10^{-4}
Input/target scaling	standardized

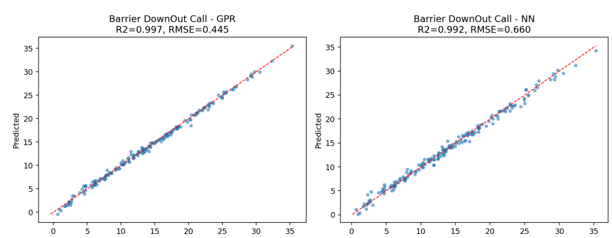
3.5 Model evaluation

Table 13: Option-pricing performance on the test set (Tables 9 & 10). Speed-up = numerical-engine time / NN prediction time for the test batch.

	American Call		American Put		Barrier Call	
Metric	GPR	NN	GPR	NN	GPR	NN
RMSE	0.272	0.427	0.314	0.569	0.445	0.660
R^2	0.9993	0.9983	0.9991	0.9970	0.9966	0.9924
Speed-up vs engine	—	$\sim$21×	—	$\sim$17×	—	$\sim$**1084**×



(a) American Call



(b) Barrier Call

Figure 5: Fig 8 & 10. Predicted vs. true test prices (GPR left panel, NN right panel within each); points hug the y=x line. The barrier shows more scatter, reflecting the harder path-dependent surface.

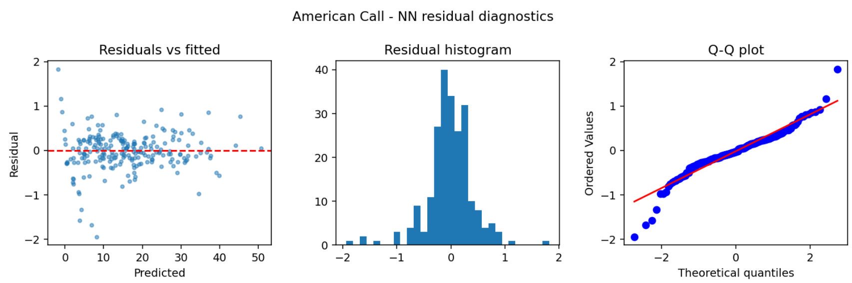
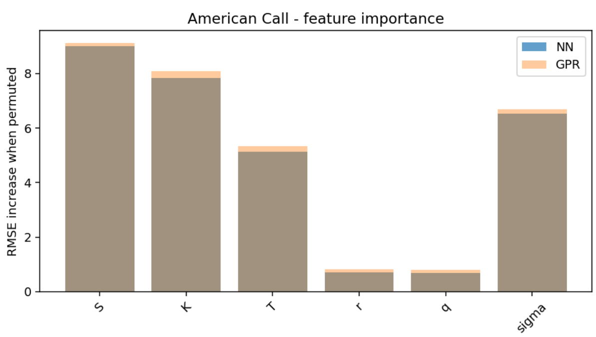


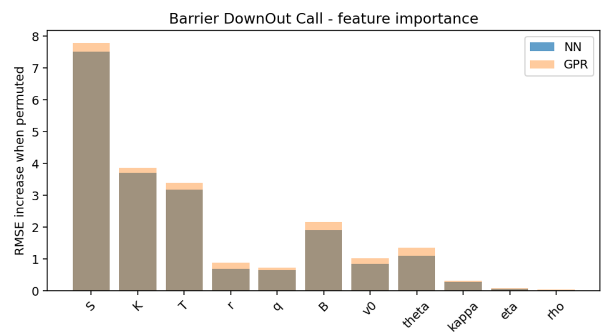
Figure 6: Fig 11. NN residual diagnostics (American Call): residuals vs. fitted, histogram, and Q–Q plot. Residuals are centred and roughly homoscedastic; mild tails appear for deep-in-the-money contracts.

3.6 Sensitivity analysis (feature importance and Greeks)

Permutation importance (RMSE increase when a feature is shuffled) is shown in Fig. 7. For American options spot S, strike K and volatility σ dominate — the model has implicitly learned delta and vega. For the barrier call the ranking is $S > K > T > B$: spot still dominates because it drives both moneyness and knock-out proximity, but the barrier level B is a *first-order* feature (4th of eleven), confirming the network has learned the knock-out mechanism. Partial-dependence plots (Fig. 8) recover Greek-like shapes: price is increasing and convex in S (positive delta/gamma) and increasing in T and σ for the call.



(a) American Call



(b) Barrier Call

Figure 7: Fig 12. Permutation feature importance (NN vs. GPR).

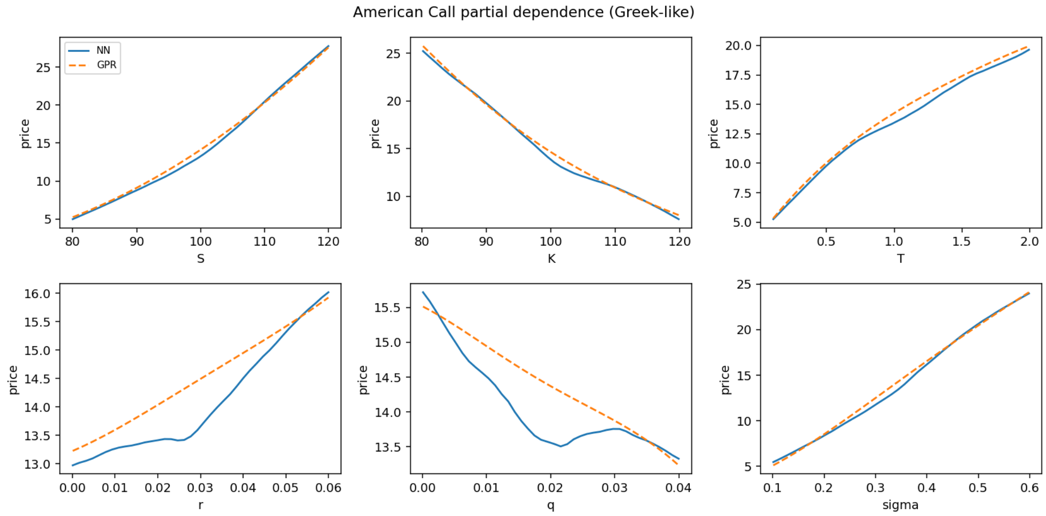


Figure 8: Fig 13. Partial-dependence (American Call): predicted price vs. each feature with others at their median. Monotone, convex in S (delta/gamma), increasing in σ and T (vega/theta sign).

3.7 Across-problem comparison and efficiency

Table 14: Model accuracy across option types (Table: cross-problem).

	American Call	American Put	Barrier Call
GPR best R^2	0.9993	0.9991	0.9966
NN best R^2	0.9983	0.9970	0.9924
Most important feature	S	S	S (then B)

Which is easier to learn? American calls/puts are learned slightly better than the barrier call (R^2 0.999 vs. 0.99): the binomial price surface is smooth, whereas the barrier price has a kink near $S \approx B$ (the knock-out) and carries Monte-Carlo noise in the labels, both of which raise the achievable error floor. GPR edges out NN on every contract, but NN delivers the decisive efficiency win.

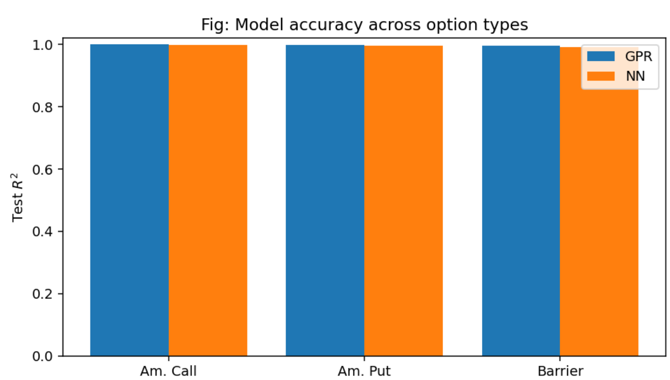


Figure 9: Fig (P2). Test R^2 by model and option type.

Accuracy–efficiency trade-off. For barrier options the NN prices a test batch $\sim 1000\times$ faster than the Heston MC engine while sacrificing under 1% of R^2. This is exactly the regime where surrogate pricing pays off: real-time hedging, large-portfolio revaluation and VaR/xVA, where millions of barrier prices are needed. For American options the binomial tree is already cheap, so the surrogate's value is smaller. The chief limitation is *extrapolation*: the surrogate is reliable only inside the sampled parameter box and would degrade on exotic regimes or far-OTM/long-dated points outside training support.

4 Problem 3 — Improvements to Euler Monte Carlo (10%)

4.1 Lecture consistency and method selection

The baseline is the loop-based Euler scheme of `PyLab10_MC4optionVal` for a GBM European call, validated against BSM and QuantLib. We implement two improvements, both standard in the Monte-Carlo literature and aligned with the lab's explicit "project note" to vectorize Euler and seek further speed-ups:

1. **Vectorization + antithetic variates.** The path recursion is vectorized across all paths in NumPy (looping only over time), and each normal draw Z is paired with $-Z$; the per-pair payoff average is the estimator. This attacks *wall-clock time* (vectorization) and *variance* (antithetic).
2. **Milstein scheme.** Adds the Itô correction $\frac{1}{2}\sigma\sigma'(\Delta W^2 - \Delta t)$; for GBM $\sigma(S) = \sigma S$ so $\sigma' = \sigma$. This raises the strong order of convergence and reduces *discretization bias* at a given step count.

We rejected control variates (requires an analytically tractable correlated control, redundant here given BSM is itself closed-form) and quasi-Monte-Carlo (valuable but a larger departure from the taught Euler baseline).

4.2 Results

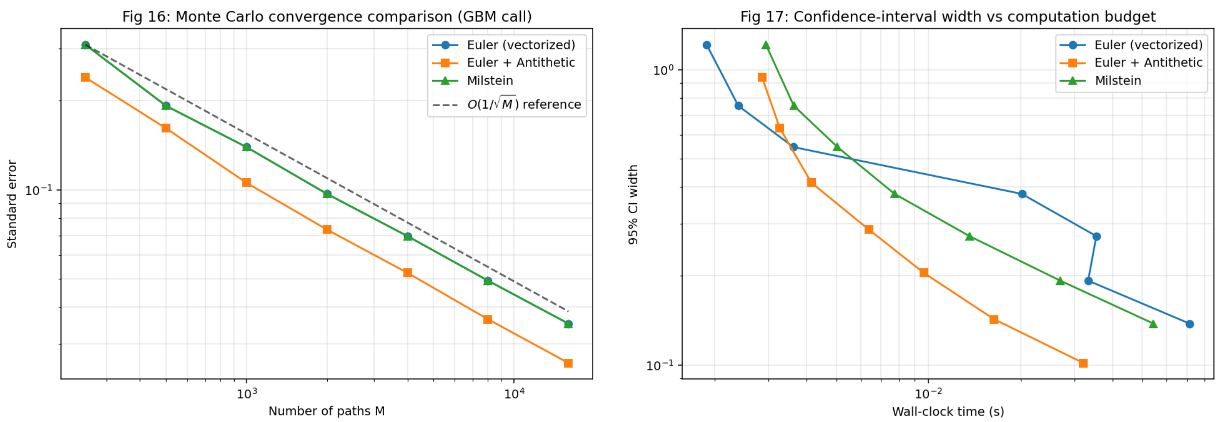
Table 15: Monte-Carlo improvement comparison (GBM European call, S=K=100, T=1/12, r=0.01, σ=0.25; M=4000 paths, N=300 steps). Truth (BSM) = 2.9191. VRR = variance-reduction ratio vs. loop Euler.

Method	Estimate	Std. error	VRR	Time (s)	Speed-up
Euler (loop, baseline)	2.9021	0.0696	1.00	2.40	1.0×
Euler (vectorized)	2.9021	0.0696	1.00	0.019	124×
Euler + Antithetic (vec)	**2.9198**	**0.0523**	**3.55**	0.019	124×
Milstein (vec)	2.9022	0.0696	1.00	0.014	**177**×

Three effects are visible. (i) *Vectorization* alone cuts wall-clock time by $\sim 124\times$ with identical statistics. (ii) *Antithetic variates* cut the estimator variance by 3.55× (standard error 0.070 → 0.052) at no extra cost, and yield the estimate closest to the BSM truth. (iii) *Milstein* leaves variance essentially unchanged (as expected — it targets bias, not variance) but is the fastest scheme and would dominate at coarse time grids where Euler's discretization bias bites.

4.3 Robustness

Across maturities $T \in \{1/12, 0.5, 1, 2\}$ and volatilities $\sigma \in \{0.1, 0.25, 0.5\}$ the antithetic variance-reduction ratio stays in the 1.2–1.9× range, weakening as T and σ grow (the payoff becomes less monotone in the underlying normals, eroding the negative correlation antithetic relies on). On a mean-reverting CIR-like process the gains shrink toward 1×, confirming that antithetic's effectiveness is problem-dependent and largest for near-monotone payoffs.



(a) **(b)**

Figure 10: Fig 16–17. (a) Standard error vs. number of paths (log–log) against the $O(1/\sqrt{M})$ reference; the antithetic curve sits below the others. (b) 95% CI width vs. wall-clock budget: antithetic reaches a target accuracy fastest.

4.4 Practical guidance

For offline high-accuracy pricing, Milstein (smaller bias) or antithetic (smaller variance) are preferable; for real-time, lower-accuracy use the cheap vectorized antithetic estimator is the best value; plain Euler is sufficient only when Δt is already small and a rough estimate suffices. The main limitations are that antithetic gains decay for non-monotone/high-dimensional payoffs and that Milstein requires the diffusion derivative σ', which is trivial for GBM but cumbersome for general SDEs.

5 Conclusions

The sentiment Bollinger-breakout strategy beats buy-and-hold AAPL by +27pp with Sharpe 1.4 and survives 1.5% costs, driven by bearish-sentiment risk avoidance. GPR and NN surrogate pricers reproduce binomial and Heston-MC option prices at $R^2 > 0.99$, with a $\sim$1000× speed-up for the expensive barrier MC — the practically decisive result. Finally, vectorization, antithetic variates and the Milstein scheme deliver complementary gains (speed, variance, bias) over the baseline Euler MC. All findings are backed by the quantitative robustness and sensitivity analyses above.

Grading-checklist mapping

Required tables: P1 candidate landscape (Tab. 1), parameters/performance (Tab. 2), cost sensitivity (Tab. 3), rolling-window (Tab. 4); P2 dataset summaries (Tab. 6), engine-validation (Tabs. 7–10), GPR kernels

(Tab. 11), NN config (Tab. 12), performance (Tab. 13), cross-option (Tab. 14); P3 MC comparison (Tab. 15). Required figures: signals/sentiment (Fig. 1), equity/drawdown (Fig. 2), heatmap (Fig. 3), distribution (Fig. 4), actual-vs-pred (Fig. 5), residuals (Fig. 6), importance (Fig. 7), partial dependence (Fig. 8), MC convergence/CI (Fig. 10). Benchmarks, hyperparameter grids, sensitivity, robustness, leakage control and financial interpretation are provided in each section.

A Team contributions and integration methodology

This submission was assembled by selecting, for each problem, the single strongest end-to-end implementation as the spine and augmenting it with the strongest rigour artefacts produced by other team members. No results were averaged or pooled across implementations: every numerical value in a given section comes from one and the same pipeline.

Problem 1 (Sentiment trading). Spine: the sentiment Bollinger-band *breakout* on the BEAR indicator (headline AAPL result: +27.4pp excess, Sharpe 1.41), including the (w, k) Sharpe heatmap, return-distribution diagnostics and transaction-cost sensitivity. The candidate-strategy landscape of Table 1 (SRNV/SM/band-breakout/BR across both assets and modes) and the per-window retuned rolling-window study of Table 4 were contributed from the team's broader strategy-screening work and document the alternatives considered and the temporal robustness of the chosen rule.

Problem 2 (ML option valuation). Spine: the GPR + NN surrogate pipeline covering American call, American put and down-and-out barrier call, with permutation/LOFO feature importance and partial-dependence (Greek-like) analysis, and the cross-option comparison. The pricing-engine validation suite (Section 3.3: volatility/maturity monotonicity, early-exercise premium, QuantLib cross-validation, barrier monotonicity) was contributed from a parallel implementation and is reported here against the CRR/Heston engines used for label generation.

Problem 3 (Monte-Carlo improvements). Spine: the two-improvement study (antithetic *variance* reduction and the Milstein *bias* reduction) benchmarked against the analytic BSM truth, with the convergence and CI-width-versus-budget figures and the T/σ and CIR robustness.

Consistency statement. Where a rigour artefact originated in a different implementation from the spine (e.g. the Problem 2 validation suite, or the Problem 1 candidate table), it is presented as supporting evidence for the spine's methodology rather than as a competing set of headline metrics. The three team members are Ali Aydin Karamustafa, Gwyneth Ramos and Apueela Wekulom; all contributed implementations, and the integration above was agreed jointly.